A Divide-Consistent, Seamless QGIS Workflow for SOTA Prominence (150 m) in Austria with Post-Computation Review Layers

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***Abstract*—Summits on the Air (SOTA) certification in Austria requires a minimum topographic prominence of 150 m. We present an automated, reproducible workflow that computes national prominence directly on a raster digital terrain model. The core contribution is *PixelMinimax*, a pixel-level descending Union-Find algorithm: DEM pixels are processed from highest to lowest elevation, and the key col of each summit is assigned at the moment its component first merges with a higher-elevation component. We prove that this first merge occurs exactly at the key-col elevation (Theorem 1), so the method requires neither an intermediate basin-saddle graph nor parameter tuning, and it correctly handles mountain-to-plain transition zones. The $O(n \log n)$ computation runs at 10 m resolution and is followed by a dual 1 m refinement of summit and key-col elevations for candidates within an empirically conservative ±20 m band around the threshold. Applied to the national Austrian ALS-derived DEM (2 450 224 128 pixels), the workflow identifies 2433 qualifying summits and agrees with 95.4 % of the official SOTA Austria database in a cross-comparison. We analyse the remaining discrepancies—362 newly computed candidates and 100 uncomputed database entries—and attribute them to identifiable geometric and data-coverage causes.**

***Index Terms*—ALS DEM, Austria, divide-consistent, GIS workflow, GRASS GIS, QGIS, SAGA GIS, SOTA, topographic prominence, Union-Find**

I. Introduction

Topographic prominence is the elevation difference between a summit and its *key col* on the optimal path to higher terrain. Let P_i denote a summit, let $z(x)$ denote terrain elevation, and let $H_i = \{x : z(x) > z(P_i)\}$ denote the terrain higher than P_i. The key-col elevation is the maximin connection elevation

$$z(K(P_i)) = \max_{\gamma \in \Gamma(P_i, H_i)} \min_{x \in \gamma} z(x), \tag{1}$$

where $\Gamma(P_i, H_i)$ is the set of admissible paths from P_i to higher terrain. The prominence of P_i is then

$$\rho(P_i) = z(P_i) - z(K(P_i)). \tag{2}$$

If several saddles attain the same maximin elevation, any one of them may serve as the representative key col $K(P_i)$; the prominence depends only on $z(K(P_i))$.

In the implementation sections, $k[P_i]$ denotes the stored key-col elevation for summit P_i; it is initialized as NaN and, once assigned, equals $z(K(P_i))$.

Beyond amateur radio activities such as Summits on the Air (SOTA), robust prominence calculation has applications in geomorphology, hydrological modelling, and peak-bagging databases [1]. Local-window or naive watershed methods can miss the true key col and misestimate prominence, as illustrated in Fig. 1. Raster-window tools such as GRASS `r.param.scale` are unsuitable for SOTA certification because their fixed search radius cannot locate key cols that lie outside it.

This paper presents an automated and reproducible system designed for national-level certification. The central contribution is a novel pixel-level descending Union-Find algorithm that directly implements the water-analogy definition of prominence and addresses inherent limitations of basin-graph approaches in mountain-to-plain transition zones. The workflow scales to Austria's national LiDAR DEM via a seamless 10 m computation paired with a targeted dual 1 m refinement.

Intuitively, the method imagines the water level dropping across the whole terrain at once. Each summit starts as its own island; as the water falls, islands grow and eventually touch. The elevation at which a summit's island first joins a taller island is exactly its key col, and the height it has stood alone above that point is its prominence. Doing this for every pixel simultaneously—rather than searching a fixed window around each peak—is what makes the result consistent across an entire country.

A. Contributions

The main contributions of this work are:

- **PixelMinimax**, a pixel-level descending Union-Find algorithm that computes topographic prominence directly on a raster DEM, without constructing an intermediate basin-saddle or divide-tree graph;
- a proof (Theorem 1) that a summit's component first merges with higher terrain exactly at its key col, together with the NaN-guard invariant that enforces this during the descending scan;
- an $O(n \log n)$ characterisation with an out-of-core implementation demonstrated on a national 10 m DEM of 2 450 224 128 pixels;
- a dual-resolution (10 m→1 m) refinement scheme with an ±20 m decision band around the 150 m threshold that is empirically conservative for the Austrian production run; and
- a national-scale cross-comparison against the official SOTA Austria database, including a structured analysis of the observed discrepancies.

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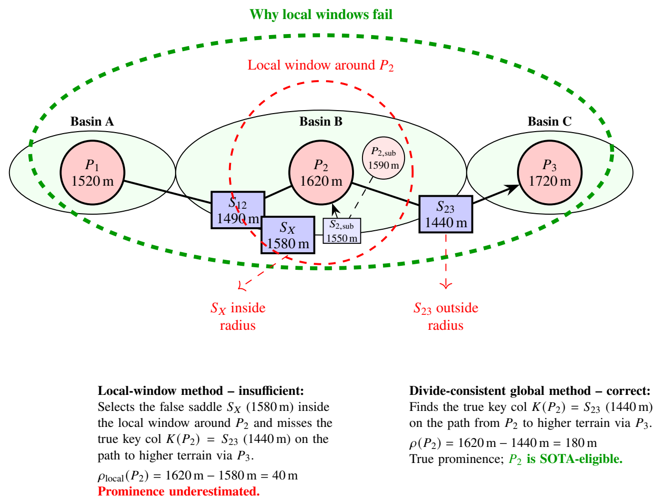


Figure 1. A local-window method can select the false saddle S_X (1580 m) inside the inspection window around P_2 and miss the true key col $K(P_2) = S_{23}$ (1440 m) on the path to higher terrain via P_3. This yields an incorrect local estimate $\rho_{\text{local}}(P_2) = 40$ m. A divide-consistent global method correctly assigns $K(P_2) = S_{23}$, yielding $\rho(P_2) = 180$ m and making P_2 SOTA-eligible. The same diagram also illustrates an intra-basin subsidiary case: for $P_{2,\text{sub}}$ (1590 m), the nearest higher terrain is P_2 within the same basin, so the key col is $S_{2,\text{sub}}$ (1550 m) and the local prominence is only 40 m.

II. Related Work

The calculation of topographic prominence has evolved through several major paradigms, summarised in Table I.

A. WinProm and Semi-Automatic Divide-Tree Logic

WinProm is one of the first semi-automatic implementations of divide-tree logic for prominence calculation [1], [2]. Its limitations include reliance on manual saddle correction, sensitivity to tile-border handling, and pruning heuristics that may discard candidate peaks.

B. Global Divide-Tree Methods

Andrew Kirmse developed a fully automated, global-scale implementation of the divide-tree algorithm [3], [4] using high-performance C++ tools. The method performs uphill walks from each saddle to identify the two peaks it connects, then resolves topological cycles. While a computational benchmark against these specialised tools was outside the scope of the present study, the approaches differ fundamentally in focus: Kirmse's method is optimised for continental-scale cataloguing, whereas this workflow is designed for national-level, reproducible certification within an open-source GIS environment.

C. Other GIS-Based Landform Analysis Methods

Methods such as the Topographic Position Index (TPI) [5] and Geomorphons [6] operate on fixed-size neighbourhoods and are therefore unable to locate key cols that may lie at large distances from the summit. The present work addresses this specifically non-local search problem.

D. Positioning This Contribution

This workflow prioritises auditability and reproducibility within a standard open-source GIS ecosystem. The pixel-level Union-Find algorithm addresses a known failure mode of basin-graph approaches in transition zones while retaining full QGIS auditability through the published intermediate hydrological layers. Because QGIS and its companion tools (GRASS, GDAL, SAGA) are free and open-source software, the entire pipeline can be reproduced at no licensing cost, and its tight integration with PostgreSQL/PostGIS and Python makes the auditable intermediate layers and the scripted processing steps straightforward to inspect, extend, and re-run on independent hardware.

III. Methods

A. Workflow Overview

The pipeline (Fig. 2) proceeds through the following stages, numbered as in the production scripts: data preparation and 40 km cross-border padding (Step 0); seamless hydrological conditioning (SAGA fill-sinks, GRASS `r.watershed`) to produce auditability layers (Step 1); pixel-level descending Union-Find (PixelMinimax) directly on the padded 10 m DEM for key-col identification, recording both key-col elevation and coordinate (Step 2, with routing-target resolution in Step 2b); dual 1 m refinement of summit and key-col elevations for borderline candidates, with final filtering for $\rho \geq 150$ m (Step 3); name enrichment with BEV DLM Geonamen and Bundesland join (Step 4); cross-validation against the official SOTA Austria database with status-coded output and direct diagnostic attribution (Step 5, refined by the diagnostic chain of Steps 5b–5d); and post-validation key-col visualisation export for QGIS (Step 4b). In this sequence, SOTA acts only as a comparison layer. The Austrian border, Bundesland, DEM, and naming layers remain the authoritative primary inputs for the computation itself.

A practical refinement of the production workflow is that all runs are now aligned to a *fixed national* 10 m *grid* in EPSG:25833. Step 0 therefore uses `gdalwarp -tr 10 10 -tap`, so that the Austria-wide run and any derived Bundesland dry-runs share the same target raster lattice. This reduces avoidable summit/key-col coordinate differences caused purely by shifted raster origins and improves the comparability of local validation runs against the authoritative national computation.

In the current production workflow, Step 1 (SeamlessHydrology) remains an auditability layer and is not required by the prominence computation (Step 2) or any downstream stage.

OpenStreetMap peak names are not inserted into Step 4 and are not used by Step 5d. If integrated in QGIS, they form an optional Step 5e-N1 evidence layer for missing BEV names only. This keeps authoritative BEV name assignment, manual review, and future database-name corrections as separate auditable stages.

Table I
Methodological Comparison of Prominence Calculation Approaches.

Aspect	WinProm (Semi-automatic)	Global Divide-Tree (Kirmse)	This Workflow
Automation	Requires operator review and edits, especially near tile edges and for ambiguous saddles.	Fully automated; designed for large-scale global cataloguing.	Prominence computation fully automated, no manual edits; database reconciliation is a separate structured review.
Core Method	Topological divide-tree logic with peak-saddle graphs.	Peak-centric topological method; connects peaks via saddles using uphill walks.	Pixel-level descending Union-Find on the national DEM; directly implements the water-analogy without an intermediate graph.
Conditioning	Implicit; often operates on unconditioned DEMs.	Dependent on input DEM quality.	Explicit hydrological conditioning (SAGA Wang-Liu fill) with documented parameters.
Saddle Identification	Topological; subject to errors at tile borders.	Saddles link peaks along ridges; edges matched at tile boundaries.	Natural emergence: the key col is the elevation at which two components first merge during descending pixel processing. Drainage outlets are handled correctly via processing-order priority.
Key-Col Search	Parent linkage via divide tree; may miss distant cols.	Fully automated via a merged, pruned divide tree.	Pixel-level Union-Find; key col assigned at first contact with a higher-elevation component. A NaN-guard prevents overwriting by later, lower-elevation connections.
Scale	Regional to national; requires manual oversight.	Global/continental; uses high-performance C++ tools.	National; designed for 64 GB+ workstations running a seamless national computation.
Auditability	Limited; manual steps hinder traceability.	High; open-source code with intermediate outputs.	High; all intermediate layers (`basins_nat`, etc.) are published for direct QGIS inspection.
Tile Borders	Prone to mismatches requiring manual correction.	Edge-handling logic; some border effects remain possible.	No tiles used. Seamless national run on a 40 km buffered DEM substantially reduces edge artifacts.
Resolution	Fixed input resolution; manual checks for critical cases.	Optimised for single-resolution, continental-scale DEMs.	10 m national compute with dual targeted 1 m refinement of both summit and key col elevations near the 150 m threshold.
Certification	Variable; manual steps reduce auditability.	High for global lists.	Designed specifically for threshold-based certification with transparent, repeatable decisions.

B. Data and Coordinate Reference System

Input datasets from Austria's Federal Office of Metrology and Surveying (BEV):

- **DTM:** ALS-derived 1 m bare-earth digital terrain model (DGM), release 2024 [7]. Native CRS: ETRS89-LAEA Europe (EPSG:3035).
- **Boundaries:** VGD national border and state polygons, version 2025 [8]. Native CRS: MGI / Austria Lambert (EPSG:31287).
- **Names:** DLM Geonamen (points), version 2025 [9].

All processing is performed in ETRS89 / UTM zone 33N (EPSG:25833). Source layers are reprojected upon ingest. The boundary layers are the only inputs in a non-ETRS89 datum: they are held in MGI (Bessel 1841 ellipsoid, EPSG:31287), whose national realisation uses three Gauss–Krüger meridian strips. Their reprojection to ETRS89 is performed with the official BEV NTv2 grid transformation (GIS-Grid 2021, EPSG operation "MGI to ETRS89 (8)"), which is specified to better than 15 cm across Austria [10] and reports a stated accuracy of 0.14 m for this operation in the processing environment; the grid method is used in preference to a seven-parameter Helmert transform precisely because the latter leaves residuals up to the metre level. The DTM and Geonamen layers are already referenced to ETRS89 realisations and require only a projection change, not a datum shift. The $\leq$15 cm transformation uncertainty is an order of magnitude below the 5 m positional accuracy of the generalised VGD border (Section III-D2) and is therefore negligible for the border-zone and distance thresholds used here; the dominant positional terms remain the 10 m rasterisation and the VGD generalisation, not the datum transformation. All distance-based operations in the workflow (buffers, nearest-neighbour searches, border distances, line lengths, and summit–key-col offsets) therefore operate in a consistent metric coordinate system. This is methodologically essential, because even small CRS inconsistencies would otherwise propagate directly into distance thresholds, audit statistics, and border-zone classifications.

C. Coordinate and Summit-Identity Validation

The prominence computation identifies summit/key-col systems, but the final database geometry requires an additional validation layer. The workflow therefore introduces Step 5d, which compares the legacy SOTA database coordinate, the BEV-NAMEN reference point, and the calculated DEM peak against the ALS 1 m DEM. For each point, the elevation at the coordinate and the local 1 m maximum in its neighbourhood are recorded.

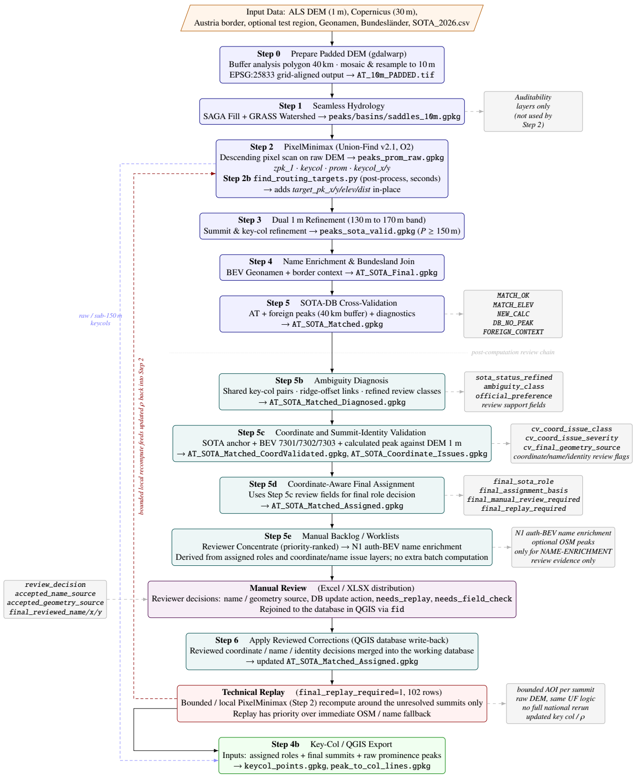


Figure 2. Pipeline overview. Hydrological conditioning (Step 1) produces auditability layers and summit candidates. Prominence is computed in Step 2 by the pixel-level PixelMinimax algorithm operating directly on the padded 10 m DEM, without requiring a basin-saddle graph. Post-validation key-col visualisation export is performed after the database cross-check.

This distinction is important because three different problems can occur. First, the SOTA database coordinate may be shifted away from the actual summit while the summit identity and height are still plausible. Second, the BEV-NAMEN point is an official reference but may not always lie exactly on the local highest cell. Third, the calculated peak can be slightly offset from the local 1 m maximum or can identify a higher neighbouring point on the same ridge without necessarily defining the official named summit.

Step 5d is therefore not a post-hoc modification of the prominence algorithm. It is a coordinate and summit-identity audit used to decide whether the final geometry should come from the SOTA database, the BEV-NAMEN point, the calculated peak, or manual review.

D. Input Data Accuracy and Resolution Constraints

The workflow relies on three BEV input layers whose intrinsic accuracy and the computational resampling step impose distinct constraints on the output. These limits are independent of the algorithm and should be understood before interpreting any result. For the Austrian production run, all primary inputs are official base datasets: the BEV ALS-derived DEM, the BEV DLM Geonamen layer, the katastralgemeinden-sharp Austrian national border, and the katastralgemeinden-sharp Bundesland polygons, supplemented by Copernicus GLO-30 only outside ALS coverage. The SOTA database is not a primary input source; it is used solely for external comparison and consistency checking.

1) ALS DEM: Vertical Accuracy vs. Computational Resolution: The national elevation source is the BEV ALS-derived 1 m *digital terrain model* (DTM, German *Digitales Geländehöhenmodell*, DGM): a bare-earth surface from which vegetation and buildings have been removed during ALS classification, derived from the corresponding 1 m digital surface model (DSM) [7]. It carries a vertical accuracy of ±0.50 m, with larger deviations possible in high-mountain terrain, making it the most precise elevation source available for Austrian territory. Being bare earth, the 1 m DTM does *not* contain trees, masts or buildings as spurious summits; the residual high-frequency variation it does carry stems from individual boulders, micro-relief and filter / interpolation artefacts of the bare-earth classification, which can still produce spurious local maxima at 1 m resolution. For national-scale processing, however, the full 1 m raster ($\approx 8.4 \times 10^{10}$ pixels for Austria) cannot be held in RAM; the DTM is therefore resampled to 10 m by averaging 10×10 ALS pixels into one output pixel. This resampling step introduces two effects that are distinct from the original measurement error:

- **Elevation smoothing:** Sharp summit spires and narrow col throats are averaged with lower-elevation neighbours, systematically compressing computed prominence. The dual 1 m refinement (Section III-J) corrects this for borderline candidates. Conversely, the same averaging suppresses the 1 m bare-earth noise noted above, removing spurious single-pixel maxima.
- **Horizontal displacement:** The centre of a 10 m pixel may lie up to 5 m from the true highest (or lowest) point within that cell. At national borders, this displacement determines whether a border-straddling summit pixel is assigned to the Austrian or the foreign DEM sector—irrespective of how precisely the original ALS measurement located the summit. The `border_dist_m` output field identifies peaks within this uncertainty band.

The 1 m ALS precision is therefore *preserved only within the dual refinement step*; outside that band the effective horizontal localization uncertainty is ±5 m (half the 10 m cell).

2) VGD National Border: Positional Accuracy: The BEV Verwaltungsgrenzen (VGD) national border polyline is geometrically generalised at a scale of 1:50 000 and carries a positional accuracy of ±5 m [8]. Combined with the ±5 m DEM rasterisation displacement, the effective uncertainty zone in which a peak's country attribution cannot be determined from raster data alone is on the order of ±7–10 m either side of the nominal border line (root-sum-square to linear combination of the two ±5 m terms). Peaks within this zone receive `border_zone = BORDER_ZONE` in the output and require manual verification. If the summit pixel falls beyond the border line (`border_dist_m < 0`, `border_zone = OUTER`), the peak is excluded from the Austrian computation and may appear as `DB_NO_PEAK` for reasons of pixel assignment rather than genuine prominence deficiency.

3) DLM Geonamen: Name Matching Radius: The BEV DLM Geonamen point layer records official summit names. The DLM is in principle scale-free; each name carries an

Erfassungsart (capture-method) attribute that falls into one of three accuracy tiers—ALS (highest, but rare for summit names), digital orthophoto (DOP, $\approx$20 cm resolution), and the KM50 national map series (the usual case)—from which the positional accuracy of the point can be derived [9]. In practice this yields a positional offset of approximately $\pm$10–30 m of the Geonamen point relative to the ALS-derived summit coordinate, because the underlying names were captured at KM50/DOP accuracy and need not coincide with the highest DEM pixel. The workflow uses a 30 m search radius for name matching (Table II), which was chosen to cover this offset; the `name_dist_m` output field records the actual distance for each match. Summits unmatched within this radius receive `name_match = 0`; the 30 m radius achieves 85.3 % coverage in the national production run (Section IV-D).

E. Processing Strategy

*1) DEM Resolution (*10 m *vs.* 1 m*):* Processing the full national 1 m DEM hydrologically exceeds the RAM of typical workstations. Aggregating to 10 m makes the problem computationally tractable: the national PixelMinimax run is executed out-of-core, processing the padded 10 m DEM with an external merge sort and SSD-backed memory-mapped arrays on a 64 GB workstation with fast NVMe scratch storage. The 1 m DEM is not discarded but reserved for the targeted refinement of borderline candidates described in Section III-J, which recovers the sub-metre accuracy where it affects certification.

2) Cross-Border Consistency: A 40 km external buffer of Copernicus DEM terrain is applied before analysis [11]. Clipping to a political border introduces artificial sinks and walls that bias the watershed structure and can cause key cols in border regions to be missed or mislocated.

3) Fixed National Grid for Local Comparability: Because SOTA certification is defined nationally, reproducibility is not only a matter of algorithmic logic but also of raster alignment. In the current production workflow, Step 0 aligns all runs to a common national 10 m lattice in EPSG:25833 using `gdalwarp -tr 10 10 -tap`. The parameter `-austria-border` always refers to the true Austrian national border, while `-test-region` optionally defines a local analysis polygon for debugging or dry-runs.

This distinction is important. Without a fixed target lattice, a local Bundesland run and the Austria-wide run may produce summit or key-col coordinate differences of a few metres even when the same physical summit is identified. Such differences arise from shifted 10 m pixel centres rather than from a change in the underlying prominence logic. Snapping all runs to the same national grid removes this avoidable source of geometric variation.

A fixed national grid does *not* imply that local runs become fully authoritative substitutes for the national run. PixelMinimax remains a global method on the buffered DEM, so differences may still occur where the relevant topographic context lies outside the buffered local extent or where the summit itself falls in the border uncertainty zone.

F. Environment

QGIS 3.34+ [12] with GRASS GIS 8.x [13], GDAL 3.8+ [14], and SAGA GIS 9.x on Apple Silicon macOS. The national production run reported here used an external NVMe SSD as scratch storage for QGIS temporary files and the out-of-core PixelMinimax stage.

G. Hydrological Conditioning and Auditability Layers

The national 10 m padded DEM is processed as a single continuous surface. The resulting layers serve two purposes: (a) providing candidate summit locations for clipping to the study area; and (b) enabling visual inspection of intermediate GIS products in QGIS, which supports the auditability requirements of official certification. These layers are not used directly for prominence computation, which is performed independently by PixelMinimax (Section III-H).

The pipeline supports `-skip-step1` (bypass Step 1 entirely with placeholder files) and `-only-step1` (run Step 0 and Step 1 in isolation for auditability review, then exit without touching downstream outputs). Both flags exploit the fact that SeamlessHydrology is not consumed by PixelMinimax; Step 2 reads the padded DEM directly. SeamlessHydrology writes empty placeholder files for the auditability layers, and proceeds directly to PixelMinimax. This is the documented production configuration for national runs where the SAGA densification step exhausts available RAM; it has no effect on the prominence computation, which reads only the padded 10 m DEM.

At full national scale, the auditability saddle extraction remains technically more demanding than the prominence computation itself because it densifies and samples basin divides. In the current production state, the Austria results reported in this paper are therefore based on the documented production path consisting of Step 0 and Steps 2–5d (including the Step 4b export), while the Step 1 audit layers remain a separate engineering target for fully robust national generation.

1) **Fill Sinks:** SAGA `fillsinkswangliu` creates a hydrologically conditioned DEM [15], [16]. A minimum slope of 0.01 enforces connectivity and prevents artificial flat areas.
2) **Delineate Watersheds:** GRASS `r.watershed` with a threshold of 5000 cells partitions the landscape into catchment basins [17], [18]. The resulting basin polygons are published as an auditability layer.
3) **Identify Peak Candidates:** Local topographic maxima on the original (unfilled) 10 m DEM are identified with a 3×3 focal maximum on the raw DEM. Plateau regions are polygonised and reduced to representative centroids for the auditability output `peaks_nat`. PixelMinimax independently identifies local maxima across the full DEM raster and does not rely on this per-basin filtering.
4) **Extract Saddles (Auditability):** Saddle elevations along basin boundaries are extracted by sampling the filled DEM at densified (10 m interval) divide vertices. These are stored as `saddles_nat` for QGIS inspection; they are not consumed by the prominence algorithm.

H. Pixel-Level Descending Union-Find (PixelMinimax)

1) Motivation: An alternative formulation builds a basin-saddle graph where each edge weight is the minimum elevation along the shared basin boundary. In purely alpine terrain, this minimum corresponds to the topographic saddle. In mountain-to-plain transition zones, however, basin boundaries cross valley floors at drainage outlets, and the boundary minimum is then a near-valley elevation rather than a ridge saddle. No reliable heuristic can distinguish a drainage outlet from a topographic saddle in a 10 m DEM, because both can appear as local elevation maxima in the flow field. The pixel-level approach avoids this problem through processing order.

2) Algorithm: All valid DEM pixels are sorted in descending elevation order and processed sequentially. A Union-Find data structure maintains connected components of already-processed pixels. For each pixel p at elevation $h = z(p)$:

1) Mark p as processed and union it with each already-processed 8-neighbour that belongs to a different component.
2) When two components C_A and C_B, with highest summits P_A and P_B satisfying $z(P_B) > z(P_A)$, first merge at p, assign $k[P_A] = h$ and record keycol_pos$(P_A) = \text{coord}(p)$.
3) The *first-merge invariant* (hereafter the NaN-guard) ensures this assignment is made at most once per component: any subsequent merge at a lower elevation (e.g., a drainage outlet or a plateau edge) does not overwrite an already-assigned key col. It is implemented by initialising each key col to NaN and writing it only when still unset.

Ties at equal elevation and plateau pixels are handled by processing all equal-elevation pixels before any merge is executed, so that an entire plateau is assigned to one component before its key col is determined. No-data pixels are excluded from the valid mask and do not participate in any merge.

The Union-Find uses union-by-rank with path compression. The overall time complexity is $O(n \log n)$, dominated by the initial descending sort of n valid pixels; the Union-Find operations themselves run in $O(n\,\alpha(n))$ [19], where α is the inverse Ackermann function, effectively constant in practice.

Algorithm 1 gives the pseudocode. In it, r_p and r_q denote the *representatives* (roots) of the Union-Find components containing pixels p and q; find$(\cdot)$ returns this root, and e_p, e_q are the highest-summit elevations of the two components. A merge is recorded only when the two roots differ, i.e. when two distinct components first become connected.

3) Correctness Argument: The correctness of the descending scan can be stated formally. We first fix the discrete setting. The DEM is a finite raster; adjacency is 8-connectivity (each pixel has up to eight neighbours), matching the neighbour loop of Algorithm 1. Pixels are processed in strictly descending elevation order, ties being broken by a fixed pixel index so that the order is total and deterministic. Each connected component of already-processed pixels is labelled by its highest summit. All elevations below refer to *discrete raster prominence*: prominence computed on the sampled DEM, which approximates the continuous terrain prominence up to the sampling and interpolation error discussed in Section III-D.

Algorithm 1 Descending Union-Find Key-Col Search (PixelMinimax)

```
Require: DEM array D[h][w]; valid mask; peak mask
  Sort valid pixels descending by elevation → order        ▷ O(n log n)
  Init UF with n = h × w singletons; k[·] ← NaN; keycol_pos[·] ← ∅
  for each pixel p in order do
      Mark p processed
      for each 8-neighbour q of p already processed do
          r_p ← find(p);   r_q ← find(q)
          if r_p = r_q then continue
          end if
          Let e_p, e_q be max-peak elevations of r_p, r_q   ▷ assume e_p ≥ e_q
          if e_p < e_q then
              swap(r_p, r_q);   swap(e_p, e_q)   ▷ ensure r_q is the lower component
          end if
          if isNaN(k[r_q]) then                              ▷ NaN-guard
              k[r_q] ← D[p];   keycol_pos[r_q] ← coord(p)
          end if
          union(r_p, r_q)        ▷ Higher-peak component becomes root
      end for
  end for
```

The idea is simple: scanning downward, a summit stays an isolated island until the water level reaches the *highest* pass leading to taller ground—any higher pass would already have connected them. That first connection is therefore the key col.

Theorem 1 (Key-col identification)**.** *Let P be a summit that is not the global maximum of the raster, and let q be the first pixel at which the component containing P merges, under 8-connectivity, with a component whose highest summit is higher than P. Then $z(q) = z(K(P))$; that is, q lies at the discrete key-col elevation of P.*

Proof. Write $h^\star = z(q)$ for the elevation of that first merge. We show $h^\star$ satisfies the maximin definition (1).

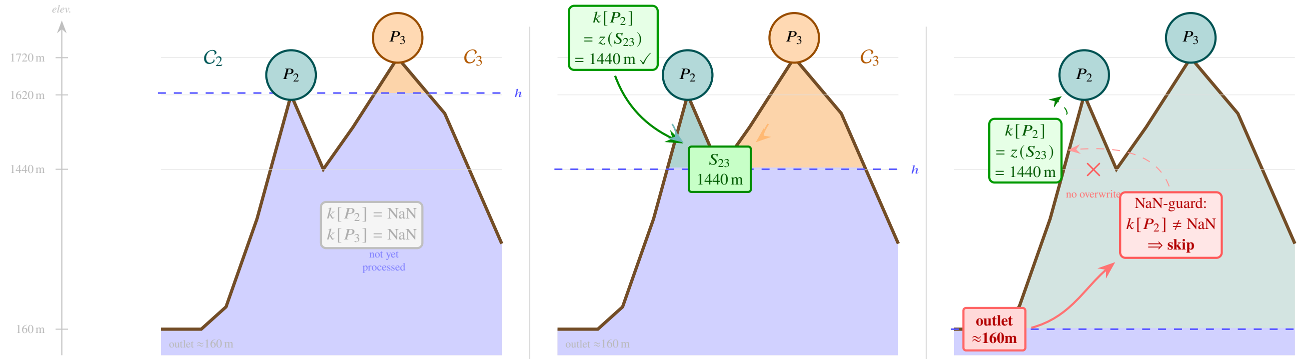
(i) A path exists at $h^\star$. During the descending scan, the component containing P consists exactly of the terrain reachable from P without dropping below the current level. At level $h^\star$ this component touches a component whose summit is higher than P. Reading off the pixels of both components at the moment they meet gives an 8-connected path from P to strictly higher terrain whose lowest pixel is q, so its minimum elevation is $h^\star$.

(ii) No path exists above $h^\star$. For every $h > h^\star$ the two components were still disjoint, so no path from P to higher terrain stays at or above h. Hence no admissible path has a minimum elevation exceeding $h^\star$.

Combining (i) and (ii),

$$h^\star = \max_{\gamma} \min_{x \in \gamma} z(x)$$

over all 8-connected paths γ from P to higher terrain, which is precisely the discrete key-col elevation $z(K(P))$ of (1).



(a) $h = 1620$ m**: two separate islands**
$k[P_2]$ = NaN; $k[P_3]$ = NaN

(b) $h = 1440$ m**:** S_{23} **merges** $C_2 \cup C_3$
$k[P_2]$ = NaN ⇒ **assign** $z(S_{23})$ = 1440 m

(c) $h \approx 160$ m**: outlet blocked by NaN-guard**
$k[P_2] \neq$ NaN ⇒ result preserved

Figure 3. PixelMinimax key-col search illustrated on a schematic terrain cross-section. Processed terrain is shaded by component (C_2 teal, C_3 orange); unprocessed terrain below the current level h is blue. (a) At $h = 1620$ m, P_2 and P_3 are separate components, $k[\cdot] =$ NaN. (b) At $h = 1440$ m, saddle pixel S_{23} merges the two components; the NaN-guard assigns $k[P_2] = z(S_{23}) = 1440$ m. (c) At $h \approx 160$ m, a drainage outlet would trigger another merge, but $k[P_2] \neq$ NaN so the NaN-guard blocks the overwrite.

(iii) Later merges cannot change the result. Every subsequent merge of P's component happens at some $h' < h^{\star}$; since P is by then already joined to higher terrain, none of them can raise the maximin value. The NaN-guard enforces this in the algorithm by refusing to overwrite a key col once assigned. □

Three boundary cases complete the statement. *Global maximum:* the highest summit of the raster never merges with higher terrain, so its key col remains NaN; by convention its prominence equals its elevation above the lowest processed pixel (or above sea level), and it is always SOTA-eligible. *Plateaus and equal-elevation ties:* because the processing order is a fixed total order, all pixels of a flat region are processed as a deterministic sequence; the first of them that produces a qualifying merge fixes the key-col elevation, and the NaN-guard prevents any later equal- or lower-elevation pixel of the same flat from altering it. The key-col *elevation* is therefore well defined, although the horizontal position of the merge pixel within a broad flat is determined only up to the plateau extent (reported with the 10 m localisation uncertainty of Section III-D). *Discreteness:* the theorem concerns raster prominence on the sampled grid; it does not claim to recover continuous terrain prominence exactly, which is why the borderline band is refined at 1 m (Section III-J).

Using the original (unfilled) DEM for this computation ensures that sink-filling artefacts do not alter saddle elevations.

4) Input DEM: PixelMinimax operates on the original (unfilled) 10 m padded DEM. The hydrologically filled DEM is used only within the SeamlessHydrology stage (Section III-G) for watershed delineation. Using the original DEM for prominence ensures that sink-filling artefacts do not alter saddle elevations. The scale of the national run (a padded 10 m DEM of 2 450 224 128 valid cells) requires the out-of-core implementation detailed in Appendix A.

5) Cross-Border Localization Uncertainty: For peaks whose key col lies within Austrian ALS coverage (the large majority of qualifying summits), the horizontal localization uncertainty is the ±5 m of the 10 m grid noted in Section III-D. For border-adjacent peaks whose key col falls within the 40 km Copernicus buffer zone, it increases to ±15 m, limited by the native 30 m Copernicus resolution. These figures reflect positional displacement, not the vertical accuracy of the underlying DEM, which for the ALS source data is ±0.50 m. The source zone of each key col is recorded in the `keycol_zone` output field.

I. Algorithm Trace: Worked Example

Tracing the algorithm for Fig. 1 with pixels S_{12} (1490 m), S_X (1580 m), and S_{23} (1440 m):

1) 1720 m: P_3 emerges, $C_3 = \{P_3\}$, $k[P_3] =$ NaN.
2) 1620 m: P_2 emerges, $C_2 = \{P_2\}$, $k[P_2] =$ NaN.
3) 1580 m: S_X processed; all neighbours belong to C_2. No cross-component merge; no assignment.
4) 1520 m: P_1 emerges, $C_1 = \{P_1\}$.
5) 1490 m: S_{12} merges C_1 ($e = 1520$ m) with C_2 ($e = 1620$ m). $k[P_1] = z(S_{12}) = 1490$ m. C_1 absorbed into C_2.
6) 1440 m: S_{23} merges C_2 ($e = 1620$ m) with C_3 ($e = 1720$ m). $k[P_2] = z(S_{23}) = 1440$ m. $\rho(P_2) = 1620\,\text{m} - 1440\,\text{m} = 180$ m.✓
7) At any lower elevation: $k[P_2] \neq$ NaN; NaN-guard prevents overwriting.

J. Dual 1 m *Refinement, Filtering, and Naming*

Candidates with prominence in the 130–170 m band undergo dual 1 m refinement.

a) Why a ±20 m *band around the* 150 m *threshold.:* The band half-width is set by the systematic prominence error of the 10 m grid rather than by rounding. As noted in Section III-D, averaging 10 × 10 ALS pixels into one 10 m cell fills narrow col throats and therefore *compresses* computed prominence: the refinement can only raise ρ, never lower it. The correct band must span the full range in which a 10 m value and its refined 1 m value can straddle the 150 m threshold. Over the national run this shift is strictly non-negative (observed range +0.4 m to +67.7 m; overall median 0 m, 95th percentile ≈ 27 m), and for the 642 candidates inside the band it averages 15.9 m — already more than 10 m. Of the 236 summits that cross from below 150 m (at 10 m) to $\rho \geq 150$ m (at 1 m), 87 start more than 10 m below the threshold, and the largest observed crossing distance is exactly 20 m. A narrow ±10 m window (140–160 m) would therefore have silently discarded those 87 valid SOTA summits by never refining them, whereas ±20 m captures every crossing case in the dataset. The upper bound is conservative for the opposite reason: because refinement never lowers ρ, no peak with a 10 m prominence ≥ 170 m falls below 150 m after refinement (0 observed), so peaks above the band need no 1 m recomputation. The symmetric band is thus the smallest window that contains all threshold-relevant cases *observed in the Austrian production run* while refining only 642 of 2433 summits rather than the full set; the bound is empirical rather than a proof that no configuration outside the band can cross the threshold.

1) Summit Elevation Refinement: A 3 km area of interest (AOI) clips the 1 m DEM around each candidate. The summit elevation is refined as the maximum within a 25 m radius of the 10 m peak coordinate, accommodating flat-topped summits and minor positional offsets [20].

2) Key Col Elevation Refinement: The PixelMinimax algorithm records the coordinate of the pixel that triggered each key col assignment (`keycol_x/y`). This pixel is the raster merge location associated with the discrete key col (Theorem 1); on plateaus or where several equivalent connections exist, its horizontal position is determined only up to the plateau extent rather than pinned to a unique geometric saddle point. The key col elevation is refined as the minimum of the 1 m DEM within a 25 m radius of that coordinate:

$$z_{\text{col,1m}} = \min_{\|\Delta\mathbf{r}\| \leq 25\,\text{m}} \left\{ \text{DEM}_{1\text{m}}(\mathbf{r}_{\text{col}} + \Delta\mathbf{r}) \right\}. \tag{3}$$

The 25 m radius accounts for the horizontal displacement between the 10 m keycol_pos coordinate and the corresponding 1 m grid, and is symmetric with the summit search radius. The refined prominence is then:

$$\rho_{\text{refined}}(P_i) = z_{\text{peak,1m}}(P_i) - z_{\text{col,1m}}(K(P_i)). \tag{4}$$

For peaks whose key col falls within the Copernicus buffer zone (outside Austrian ALS coverage), the refined prominence of (4) substitutes $z_{\text{col,10m}}$ transparently via a coverage check.

K. Post-Computation Review and Final Assignment Layers

The productive prominence engine ends with stable ALS/DEM-derived summit and key-col geometry, but the Austria production workflow now continues with a post-computation interpretation chain. Step 5 performs database matching and primary diagnostics, Step 5b exposes the review families hidden inside the broad comparison classes, Step 5c adds coordinate, name, and summit-identity diagnostics, and Step 5d assigns an explicit final summit role in `AT_SOTA_Matched_Assigned.gpkg`. This extension does not modify the core prominence computation itself; it creates a separated audit and decision layer that is necessary for ridge ambiguities, exact-position database-height conflicts, shared-key-col pair cases, and hard residuals without a near calculated partner.

1) SOTA Database Cross-Validation: Computed peaks are matched spatially against the official SOTA Austria database (filtered to `ValidTo = 31/12/2099`) within a configurable radius (default 500 m). Each feature receives a `sota_status` field:

- `MATCH_OK`: spatial and elevation agreement ($\Delta z \leq 10$ m);
- `MATCH_ELEV`: spatial match with elevation discrepancy > 10 m, which may indicate a database positioning error;
- `NEW_CALC`: qualifies by computation, absent from database—proposed for review;
- `DB_NO_PEAK`: in the database but no computed peak nearby—appended as an additional row at the database coordinates.

A colour-coded QGIS symbology file (`.qml`) accompanies the output for visual audit. After refinement, peaks with $\rho <$ 150 m are removed. Names from BEV DLM Geonamen and Bundesland polygons are joined to the final output.

L. BEV Reference Elevation and Source Comparison

The BEV DLM Geonamen layer (`NAM_7300_GELAENDEFORM_P`) carries the official summit elevation in the field `HOEHE_BODEN`. This value is the basis for Austrian topographic maps and for the SOTA Austria database, and therefore carries administrative authority regardless of its photogrammetric origin.

Step 4 (name enrichment) stores this elevation as `z_bev` (0.1 m precision) whenever a Geonamen feature is matched within the 30 m search radius. The signed difference in (5),

$$\Delta z = z_{\text{ALS}} - z_{\text{BEV}} \tag{5}$$

is stored as `z_bev_diff`, where z_{ALS} is the refined 1 m summit elevation (`z1m_max`), falling back to the 10 m raster value (`zpk_1`) if refinement was not triggered.

These fields are *reference only*: the prominence calculation uses ALS elevations exclusively for both summit and key col, maintaining a homogeneous source for numerator and denominator. A mixed computation $\rho = z_{\text{BEV}} - z_{\text{ALS,col}}$ would introduce a systematic bias whose sign and magnitude depend on the vintage of the BEV survey, and is therefore deliberately excluded.

Diagnostic use of `z_bev_diff`. Photogrammetrically derived map heights are known to lose accuracy on steep alpine slopes and at exposed summit points, and the BEV product documentation likewise notes that larger height deviations can occur in high-mountain terrain [7]. A positive Δz (ALS higher than map) is therefore expected in high-alpine terrain and reflects improved measurement technology rather than an error. A value $|\Delta z| > 5$ m warrants field verification, as it may indicate a mismatched Geonamen point or a discrepancy in the definition of the highest measurable point. The `MATCH_ELEV` status in the SOTA cross-validation layer flags entries where $|z_{\text{ALS}} - z_{\text{SOTA-DB}}| > 10$ m; `z_bev_diff` provides the topographic explanation for most such cases.

M. Outputs and Parameters

The workflow produces the qualified summit layer `AT_SOTA_Final` (prominence, key-col coordinates, BEV

Table II
PARAMETER DEFAULTS.

Setting	Default
Fill sinks (Wang–Liu) min slope	0.01
Watershed threshold	5000 cells
Peak detector method	SAGA focal maximum
Peak detector size	radius 1 cell (3×3)
Working DEM / refinement	10 m / dual targeted 1 m
Name search radius	30 m
Refinement AOI radius	3 km
Summit search radius	25 m
Key col search radius	25 m
SOTA DB match radius	500 m

name and elevation) together with the Step 5 cross-validation and review layers; the complete per-layer field schema is provided in the supplementary technical documentation. Default parameters are given in Table II with justifications in Table III.

Table III
JUSTIFICATION OF KEY WORKFLOW PARAMETERS.

Parameter	**Justification**
Fill Min. Slope *(0.01)*	Enforces hydrological connectivity; prevents artificial flat areas without excessive terrain alteration.
Watershed Threshold *(5000 cells)*	Minimum drainage area $0.5\,\mathrm{km}^2$, approximating first-order stream catchments in alpine environments. Avoids over-fragmentation while resolving distinct hydrological units.
Peak Detector Size *(radius 1 cell)*	A 3×3 neighbourhood on the raw 10 m DEM is sufficient for the auditability peak layer, while the final prominence computation is performed by PixelMinimax on the full DEM rather than by a local-window detector.
Refinement AOI *(3 km)*	Sufficient buffer to locate the true 1 m summit position accounting for 10 m/1 m positional offset, while keeping the clipped raster small.
Summit / Col Radius *(25 m)*	Covers the maximum horizontal displacement between 10 m and 1 m grids, and accommodates flat-topped summits and narrow col areas.
Name Search Radius *(30 m)*	Accounts for positional offset between computed summit coordinates and BEV Geonamen point features.

IV. RESULTS

A. National-Scale Results

The national production run used the 40 km-padded Austrian DEM and completed the documented production path consisting of PixelMinimax, dual 1 m refinement, BEV naming, SOTA database cross-validation, and post-validation key-col export. The Austrian result is based on official Austrian source layers only: the BEV ALS DEM, the BEV DLM Geonamen layer, the katastralgemeinden-sharp national border, the katastralgemeinden-sharp Bundesland polygons, and Copernicus padding only outside ALS coverage. Step 2 processed

2 450 224 128 DEM pixels. PixelMinimax produced 2677 raw peaks with $\rho \geq 130$ m, of which 886 fell within the 130–170 m dual-refinement band. Of those, 642 qualified ($\rho_{\text{ref}} \geq 150$ m) and 244 were filtered; the remaining 1791 peaks ($\rho_{\text{raw}} > 170$ m) passed directly. After the final $\rho \geq 150$ m filter, 2433 summits remained in `AT_SOTA_Final`.

Cross-validation against the official SOTA Austria database yielded

2053 `MATCH_OK` summits,

18 `MATCH_ELEV` cases,

362 `NEW_CALC` candidates, and

100 `DB_NO_PEAK` entries, corresponding to an Austrian database coverage of 95.4 % (2071/2171 Austrian database entries). Federal-state attribution uses a direct spatial join against official Bundesland polygons. Step 4b exported 2433 qualifying key-col objects plus 20 `DB_NO_PEAK` saddles recoverable from the raw PixelMinimax layer, for a total of 2453 visualisation objects (80 further `DB_NO_PEAK` rows had no exportable saddle).

Step 5b (ambiguity diagnosis) refines the broad Step 5 status classes into review-relevant structures; the national comparison totals are reported in Section IV-B and Table IV. The subsequent post-validation layers (Steps 5c and 5d) add coordinate, name, and identity diagnostics and assign explicit final roles *without altering any prominence, key-col, saddle, or matching value*. In summary, the coordinate-aware final assignment confirms `final_sota_flag`=1 for 2124 summits and flags 681 rows for manual adjudication; the remaining rows are foreign-context features outside the national boundary. The full per-flag breakdown is reported in Appendix B to keep the main results focused on the prominence computation and its validation.

B. Cross-Comparison with the Official SOTA Austria Database

The workflow was cross-compared against the official SOTA Austria database at national scale. The database is a curated, incrementally maintained reference rather than an independent ground truth, so the agreement figures below quantify *consistency between two independently produced summit sets*, not correctness against a verified standard. Because there is only a single national computation run, results are reported at the national level and then split by federal state. The comparison therefore rests on the agreement between the computed prominence results and the existing database entries, with the remaining discrepancies passed to the structured manual review layer.

1) National Database Comparison: Table IV summarises the national cross-validation results by federal state. The SOTA Austria database contains

2171 Austrian entries. The workflow identified 2433 summits with $\rho \geq 150$ m; 2071 (95.4 %) matched an existing database entry within the spatial and elevation tolerance (columns *DB Total*, *Matched*, *Cov.*). The remaining 100 database entries were not computed (`DB_NO_PEAK`), and the 362 `NEW_CALC` candidates are peaks computed by the workflow but absent from the current database; Step 5b classified the latter into 239 manual-review, 45 border-zone, and 12 shared-key-col cases. The causes of both groups are analysed in Section V. The spatial distribution of the resulting review workload (the priority-ranked Reviewer Concentrate) is shown in Fig. 4.

The *Final* column shows the count of summits assigned `final_sota_flag`=1 by Step 5d (confirmed SOTA summit or final summit requiring review); this figure is directly comparable to the number of certified summits the database should contain after completion of the manual review process.

C. Cross-Border Performance

Several key cols for Vorarlberg peaks were located in Switzerland, confirming that unclipped hydrology on the padded DEM is necessary for border regions. Approaches that clip the DEM to the political boundary produce artificial sinks at the border and systematically mislocate cross-border key cols.

D. Name Matching Accuracy

Geographic name matching achieved 85.3 % at the default 30 m radius in the national production run (2075/2433 matched computed summits). The BEV DLM Geonamen dataset does not cover all prominent peaks; many qualifying summits are unnamed in the authoritative data.

1) Name and Evidence Layers: Authoritative BEV names are assigned where available; 2247 of the 2433 qualifying summits (92.4 %) receive one, leaving 186 for manual name research. The authoritative-name figure exceeds the automatic 30 m Geonamen matches of Step 4 (2075) because BEV name coverage is established by the Step 5c/5e name-source analysis over the full BEV reference hierarchy (`F_CODE` 7302/7303 summit anchors), not solely by the fixed 30 m radius. The remaining 186 summits are elevations that qualify by prominence but are not categorised as named summits in the BEV inventory; for these, the SOTA-DB label is retained as a fallback reference, with optional OpenStreetMap evidence and visual verification against the official `basemap.at` map handled in the review workflow. A separate reviewer copy manages the name-source queues, the SOTA-DB fallback policy, and optional OpenStreetMap name evidence; these review-only layers do not affect prominence, key-col geometry, final roles, or replay flags, and their field-level schema is given in Appendix B.

V. Discussion

Pixel-Level vs. Basin-Graph Approaches

The primary methodological contribution is the transition from a basin-saddle graph to a pixel-level descending Union-Find. In purely alpine terrain both approaches yield comparable results, because basin boundaries coincide with ridge lines and their minimum is the topographic saddle. In mountain-to-plain transition zones, basin boundaries also cross valley floors at drainage outlets, and the graph edge minimum can then be a near-valley elevation rather than a ridge saddle. The pixel-level approach resolves this by construction: the ridge saddle is processed before any drainage outlet because it has a higher elevation, and the NaN-guard prevents later, lower-elevation connections from overwriting the result. No parameter tuning is required to handle transition zones.

Role of Hydrological Conditioning

The SAGA and GRASS preprocessing stages are retained for two reasons. First, they supply the `peaks_nat`, `basins_nat`, and `saddles_nat` layers used for QGIS visual inspection, which supports the auditability requirement of official certification. Second, the conditioned DEM underpins the watershed delineation from which summit candidates are clipped to the study area. The prominence computation itself (PixelMinimax) operates on the original unfilled DEM to avoid any saddle-elevation bias introduced by sink filling.

Dual 1 m Refinement

Earlier implementations refined only the summit elevation while retaining the 10 m key col [21]. This introduced a conservative bias: 10 m DEMs tend to raise narrow col elevations, which reduces the computed prominence. The dual refinement corrects both components for borderline candidates. Because `keycol_pos` is the exact merge pixel of the Union-Find, the 25 m zonal minimum is expected to lie at or immediately adjacent to the true topographic saddle; the radius accounts for horizontal displacement between the 10 m and 1 m grids.

Interpretation of Database Discrepancies

The cross-validation is best read not as a pass/fail test but as a structured comparison of two independently produced summit sets, whose disagreements are themselves informative. Two categories dominate.

The 362 `NEW_CALC` candidates are peaks that satisfy the computed $\rho \geq 150$ m criterion but are absent from the current database. These are not automatically errors: the database has been curated incrementally and is not guaranteed to be complete at the 150 m threshold. Their internal structure supports this reading: Step 5b classifies 239 as requiring manual review, 45 as strong border-zone candidates, and 12 as forming shared-key-col pairs with uncomputed database entries. The pronounced regional concentration—165 of the 362 fall in Tirol (Table IV)—is consistent with dense, high-relief alpine terrain where closely spaced summits near the threshold are most likely to be under-represented in a manually assembled list.

The 100 `DB_NO_PEAK` entries are database summits for which the workflow computed no qualifying peak. The two identifiable causes are (i) key cols located outside Austrian ALS coverage, i.e. on cross-border saddles resolved only at the coarser 30 m Copernicus resolution, and (ii) genuine refined prominence below 150 m, where the 1 m refinement did not lift the candidate across the threshold. Both are geometric or data-coverage effects rather than algorithmic failures, and both are surfaced—not silently discarded—through the review layers. This is why the workflow reports discrepancies for manual adjudication rather than deleting database entries automatically.

Beyond missing and additional summits, the comparison also exposes a *systematic coordinate-quality pattern* in the existing database. Of the summits examined, 911 exhibit a position conflict between the database coordinate and the computed peak; for 788 of these the independent BEV reference point confirms that the database coordinate is displaced from the local maximum, while only 73 rest on the computed peak alone. Taken together, close to 40 % of the compared database summits carry a sub-optimal coordinate. A complementary naming analysis shows that 2247 of the 2433 qualifying summits (92.4 %) receive an authoritative BEV name, leaving 186 that

Table IV
NATIONAL CROSS-VALIDATION BY FEDERAL STATE UNDER THE V5.1 DATA FREEZE. THE TABLE USES THE AUSTRIAN/TABLE-V SUBSET (`NULLIF(LAND,'NULL') IS NOT NULL`) TO AVOID MIXING FOREIGN CONTEXT ROWS WITH NATIONAL STATE TOTALS.

Bundesland	DB Tot.	Matched	DB_NO	Cov.%	NEW	Final	Conf.	P1	ManRev	Rev%	Geo	QA
Burgenland	2	2	0	100.0	1	3	2	1	1	33.3	2	0
Kärnten	270	255	15	94.4	29	261	240	21	58	19.4	105	14
Niederösterreich	197	186	11	94.4	19	187	181	6	34	15.7	107	11
Oberösterreich	200	193	7	96.5	23	197	183	14	39	17.5	59	5
Salzburg	273	268	5	98.2	51	275	233	42	91	28.1	98	7
Steiermark	446	438	8	98.2	58	442	425	17	77	15.3	124	6
Tirol	655	611	44	93.3	165	635	480	155	337	41.1	351	49
Vorarlberg	127	117	10	92.1	16	123	104	19	39	27.3	53	10
Wien	1	1	0	100.0	0	1	1	0	0	0.0	0	0
Total	2171	2071	100	95.4	362	2124	1849	275	676	26.7	899	102

Notes. The Table-V subset consists of Austrian rows with `NULLIF(land,'NULL') IS NOT NULL`; this explicitly excludes foreign/context rows whose attribute value may be the literal string `'NULL'`. *DB Tot.* = MATCH_OK + MATCH_ELEV + DB_NO_PEAK, i.e. Austrian SOTA database entries per state. *Matched* = MATCH_OK + MATCH_ELEV. *DB_NO* = DB_NO_PEAK, i.e. a database entry is present but no computed summit peak was assigned automatically, for example because the relevant saddle is cross-border, outside the run context, or because the local evidence is below the 150 m prominence threshold. *Cov.* (%) = Matched / DB Tot. *NEW* = NEW_CALC, i.e. computed candidates absent from the database. *Final* = `final_sota_flag`=1 after Step 5d coordinate-aware final assignment. *Conf.* = `FINAL_SOTA_SUMMIT`. *P1* = `FINAL_SOTA_SUMMIT_REVIEW`. *ManRev* = `final_manual_review_required`=1 within the Austrian/Table-V subset; this gives 676 rows, while the all-row total is 681. *Rev%* = ManRev / (DB Tot. + NEW) × 100. *Geo* = Step 5c coordinate/name/identity review hint, defined as `cv_coordinate_review_required`=1 OR `cv_identity_review_required`=1 OR `cv_name_review_required`=1 within the Austrian/Table-V subset; this gives 899 rows. The all-row Step 5c issue footprint is 911. *QA* = `final_replay_required`=1, i.e. the technical replay/local-recalculation backlog.

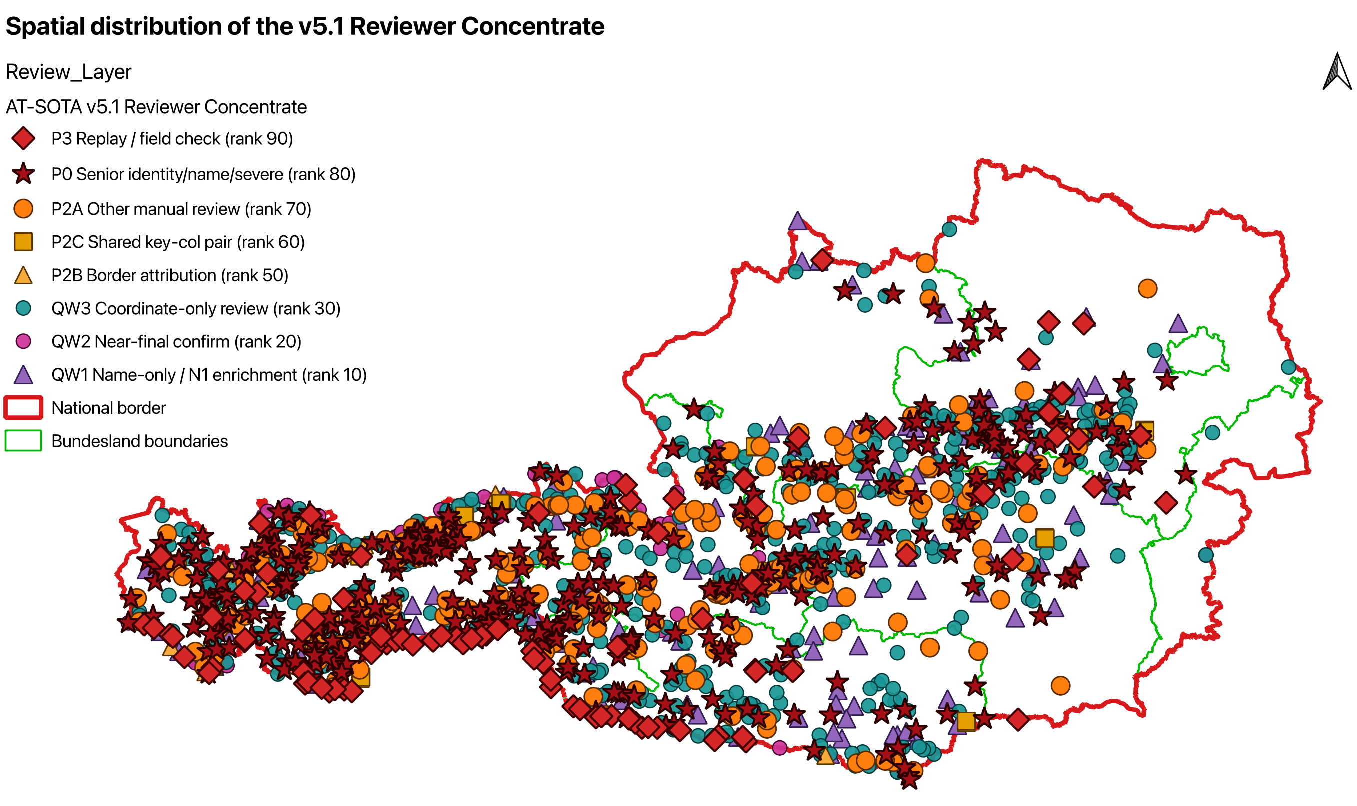


Figure 4. Spatial distribution of the v5.1 Reviewer Concentrate—1401 priority-ranked review rows (1389 of them Austrian)—over Bundesland boundaries, symbolised by review priority class from technical replay/field check (rank 90) down to name-only/N1 enrichment (rank 10). The 676 Austrian rows requiring manual review (`final_manual_review_required`=1; Table IV) are contained in this set and concentrate visibly in Tirol, while the majority of the 2433 qualifying summits carry no review flag and are omitted for legibility.

require manual name research. These findings indicate that the workflow is useful not only for detecting missing or additional peaks, but also for auditing the positional and naming quality of an existing authoritative dataset—an application that a pure prominence recomputation would not provide.

Limitations

A full 1 m national Union-Find is computationally prohibitive ($\approx 84 \times 10^9$ pixels for Austria) and may amplify noise in flat terrain [22], [23]. The 10 m national computation paired with targeted dual 1 m refinement is a deliberate design choice that balances precision and practicability.

A second class of limitations is geomorphological. On extensive *plateaus* and low-relief karst surfaces the key col can occupy a broad, near-flat region in which many pixels share the maximin elevation; Theorem 1 still identifies a valid key-col elevation, but the horizontal position of the merge pixel within such a flat is not uniquely determined and is reported with the corresponding 10 m localisation uncertainty. *Glaciated* terrain poses a distinct issue: the ALS bare-earth DTM models the ice surface at acquisition time, so summit and saddle elevations on glaciers are time-dependent and may drift between DEM vintages. Finally, residual DEM noise in flat saddles can perturb borderline prominence values; the ±20 m refinement band is designed to contain such threshold crossings, but candidates in these settings are the most likely to require manual adjudication.

The remaining source of uncertainty is the 10 m horizontal localization of key cols for peaks outside the dual-refinement band (i.e., $\rho < 130$ m or $\rho > 170$ m). For peaks well above 150 m, this has no effect on certification. For peaks close to but outside the band, the SOTA database cross-validation flags any discrepancies with the operational list for manual review.

a) Topological Boundary Cases.: Although developed and validated for Austria, the core algorithm (PixelMinimax, Steps 0–4) is territory-agnostic. Three boundary conditions are handled by construction.

First, a *single-summit territory* (no other summit in the study area exceeds the prominence threshold) produces a single all-encompassing Union-Find component; prominence is assigned correctly. Peaks from neighbouring countries that lie within the analysis buffer but outside the national boundary are handled via a dedicated `FOREIGN_CONTEXT` role; the Austrian production run contains 1 298 such foreign-context features. Importantly, a summit can be the highest within its country while still having its key col entirely within the national boundary. The Großglockner (z = 3795 m, the highest summit in Austria, ρ = 2424 m) illustrates this: the key col lies 34.4 km inside Austria at 1371 m elevation, at a Bundesland (federal state) boundary, and receives `admin_context = BL_BORDER` rather than any foreign flag. The routing from Großglockner to the next higher peak (the Ortler, outside Austria) spans approximately 91 km from the key col; the key col is the highest saddle on that traverse. A notable consequence of the output scope is that the identity of the routing-target peak is not stored in the AT SOTA dataset: the pipeline records the key col coordinates and elevation for every Austrian summit, but the foreign peak whose component triggered the merge is present only in the DEM buffer and is not exported. As of v5.1, the routing-target coordinates are resolved by a lightweight post-processing step (Step 2b) that locates, for each summit, the nearest higher peak by spatial grid search from the key-col position; implementation details and the evaluated alternatives are given in Appendix A. For the Großglockner, this identifies the Ortler ($z \approx 3905$ m, South Tyrol, Italy), approximately 91 km south-west of the key col, as the routing target.

Second, an *island state* whose land mass is fully surrounded by NoData (ocean) has no valid external merge partner in the DEM. The Union-Find stops at the last coastal pixel; prominence therefore equals summit elevation above mean sea level. This is the mathematically correct result: sea level acts as the implicit reference plane, exactly as for Everest (ρ = 8849 m), the only real-world summit whose key col genuinely lies at mean sea level because no higher peak exists anywhere on Earth. The analogy holds only for a truly isolated island with no land connection to higher terrain within the DEM; a continental summit that merely happens to be the highest in its country (like the Großglockner) does not fall into this case.

Third, a *lake island* is processed without special treatment: Step 1 assigns a constant water-surface elevation to the lake; the key col is placed at the shoreline pixels and prominence equals summit elevation above lake level.

The only Austria-specific component is the Step 5 cross-validation against the official SOTA Austria database, which can be replaced or omitted without affecting the prominence computation. The interplay between ALS measurement accuracy, 10 m resampling, and border-line precision is discussed in Section III-D; the `border_dist_m` field operationalises these constraints for per-peak audit.

A further practical limitation concerns the relationship between the authoritative Austria-wide run and local Bundesland dry-runs. Even when all runs are snapped to the same national 10 m grid, local runs remain *derived validation runs*, not replacements for the national result. Differences may still occur:

- in the `BORDER_ZONE`, where summit attribution is already uncertain at the 10 m level;
- where the relevant topographic context for the minimax path lies outside the local buffered analysis extent;
- where a local engineering workflow temporarily skips auditability-only components such as Step 1 (SeamlessHydrology).

For this reason, official Austrian SOTA decisions should always be taken from the Austria-wide run; local runs are intended for debugging, inspection, and targeted comparison.

The official SOTA Austria list provides a consistency reference rather than independent ground truth. The primary Austrian computation is based on official Austrian base datasets only; SOTA is consulted afterwards for comparison, discrepancy flagging, and audit support. In the current national production run, the Austrian database coverage is 95.4 %. Remaining discrepancies are presented transparently through the `sota_status` field and the added diagnostic review fields for further verification. A further current limitation is that fully robust national generation of the SeamlessHydrology saddle audit layer remains an engineering task; this does not affect prominence computation because PixelMinimax operates independently of the Step 1 audit layers.

Beyond SOTA Certification

The algorithm is not restricted to SOTA certification. It computes full topographic prominence for all local maxima and can be applied to any threshold-based summit inventory. Adaptation to larger geographies requires widening the refinement band to compensate for a coarser base DEM (see Appendix).

A. Post-Processing Interpretation Layer

A major methodological consequence of the Austria-wide evaluation is that broad comparison labels such as

MATCH_ELEV, NEW_CALC, and DB_NO_PEAK are not yet final summit decisions. The added Step 5b and Step 5c layers therefore belong to the audit and deployment strategy rather than the core prominence algorithm. Step 5b identifies which rows are primarily height/reference conflicts, which are ridge-like wrong-summit reviews, and which form likely shared-key-col pairs. Step 5c adds coordinate/name/identity diagnostics and Step 5d minimises manual workload by assigning explicit final roles while leaving only the hardest residuals in review or replay queues. This assignment follows a deliberately conservative confidence pattern: stable final summits carry high-confidence roles, shared-key-col pair outcomes are marked as intermediate, and low-confidence labels are concentrated in the review, unresolved, and replay classes. The design goal is to avoid false confidence in unresolved cases and to keep difficult residuals explicit rather than hiding them behind a single binary flag.

The construction of these review families and the associated reviewer worklists rests on an extensive auditable query layer rather than on opaque heuristics. In total, 95 dedicated SQL analyses were run against the matched and diagnosed GeoPackage layers to derive the case families, to cross-tabulate the coordinate/name/identity conflicts, and to assemble the priority-ranked reviewer Excel worklist. Each analysis was additionally registered as a QGIS *virtual layer*, so that its result set could be rendered directly on the map alongside the ALS DEM, the computed peaks, and the key-col lines. This dual use—SQL for the tabular logic and the same query as a live virtual layer for visual inspection—served two purposes: it provided an on-map cross-check that each family definition selected the intended summits, and it let borderline and shared-key-col cases be inspected geometrically before any reviewer decision was recorded. Because virtual layers re-execute their query against the underlying data, the visual audit always reflects the current layer state and remains fully reproducible in QGIS.

VI. Conclusion

We have presented a reproducible QGIS workflow for topographic prominence computation and SOTA certification, developed and validated for Austria but designed to generalise to any territory. The workflow:

1) implements a pixel-level descending Union-Find algorithm (PixelMinimax) that directly realises the water-analogy definition of topographic prominence and handles mountain-to-plain transition zones without parameter tuning;
2) operates on a 40 km-padded national DEM at 10 m resolution with targeted dual 1 m refinement for borderline candidates, and aligns both national and derived local runs to a fixed national 10 m EPSG:25833 grid for improved reproducibility;
3) supports a national out-of-core implementation using external merge sort and SSD-backed memory maps, while treating local Bundesland runs as validation runs rather than substitutes for the Austria-wide authoritative result;
4) cross-validates results against the official SOTA Austria database with status-coded, visually auditable output; and
5) enriches output with BEV Geonamen, Bundesland attribution, and post-validation key-col visualisation layers.

The pipeline degenerates correctly for topological boundary cases including single-summit territories and island states, as discussed in Section V.

Applied to Austria, the current production run yielded 2433 summits with refined prominence $\rho \geq 150\,\text{m}$, together with 2053 accepted MATCH_OK matches, 18 flagged elevation mismatches (MATCH_ELEV), 362 new candidates (NEW_CALC), and 100 database entries without a computed peak (DB_NO_PEAK). The validated national production path is thus operational. The remaining engineering task is fully robust generation of national SeamlessHydrology saddle audit layers, which are not required by the prominence computation itself.

The frozen v5 review state keeps computation, diagnosis, review and correction deliberately separate. Step 5c detects coordinate, name and summit-identity conflicts but does not apply database corrections. Step 5d creates final coordinate-aware roles and flags.

Step 5e turns these flags into reviewer worklists, including the BEV-first N1 name-source queue for calculated summits without BEV name coverage. These worklists are distributed to human reviewers as a priority-ranked Reviewer Concentrate (Excel/XLSX) and re-joined to the working database in QGIS through the stable fid key. SOTA-DB names are retained as existing fallback references but do not close the BEV-name question. Optional OpenStreetMap peak names may be displayed in QGIS as supplementary review evidence for N1 rows only; they do not alter v5 computation or final roles.

Step 6 then applies the reviewed coordinate/name/identity decisions and creates database-ready update exports without overwriting the original evidence layers. Only after these decisions are merged is the technical replay executed: the 102 rows with final_replay_required=1 are recomputed by a *bounded, local* PixelMinimax pass (Step 2) restricted to the neighbourhood of each unresolved summit, rather than by a full national rerun. Replay therefore has priority over immediate OSM/name fallback, and its updated key-col elevations and prominences feed back into the same output layers.

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During the preparation of this work the author used Claude (Anthropic) to assist with the implementation and optimisation of Python scripts and complex SQL queries. A portion of the Python scripts was initially developed by the author using the QGIS Model Designer and subsequently optimised with Claude. The concept, the core algorithms, and all scientific decisions, analysis, statistics, and categorisation of special cases are the author's own work. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Conflict of Interest

The author declares no conflicts of interest.

Code Availability

The production pipeline—the driver script and all QGIS Processing scripts implementing Steps 1–5d (PixelMinimax, the dual 1 m refinement, the SOTA database match, and the diagnostic and assignment stages)—will be made publicly available upon publication at https://github.com/Erwin60/sota_prominence, together with a DOI-assigned archival snapshot. The repository includes a full parameter reference and worked usage examples. The three BEV base datasets used here—the ALS-derived DGM 1 m height raster [7], the Verwaltungsgrenzen (VGD) boundaries [8], and the DLM Geonamen (Namen) layer [9]—are provided free of charge (*unentgeltlich*) by the BEV under its open-data terms and can be downloaded directly from the cited product pages, so the full input stack is publicly reproducible. The input geodata and the SOTA Austria database are not redistributed with this work and remain subject to their respective providers' terms.

Supplementary Material

The intermediate layers and visualisation styles are maintained as project materials accompanying this workflow. Archival release information (e.g. a DOI-assigned snapshot) will be added upon publication.

Appendix A
Computational Performance and Memory Estimation

The original in-RAM formulation of the national PixelMinimax run proved marginal at Austria's full padded extent. The current production implementation therefore uses an out-of-core strategy: the DEM cells are sorted via an external merge sort, while the large Union-Find index arrays are stored as SSD-backed memory maps.

a) Application to Austria: With the mandatory 40 km cross-border buffer, the padded Austrian DEM was processed as a single seamless surface. The descending sort was executed as an external merge sort, and the Union-Find state was held as SSD-backed memory-mapped `int64` arrays for parent pointers and pixel indices, together with floating-point arrays for elevation and key-col values. On an Apple Silicon workstation with 64 GB unified memory and external NVMe scratch storage, the national Step 2 run completed in 10 h 29 m 42 s; Table V consolidates the corresponding figures. Two of these figures warrant a brief explanation. Local maxima are detected with a *non-strict* 3×3 maximum filter (a pixel qualifies if it is not lower than any of its eight neighbours), so every pixel of an equal-elevation plateau counts; flat terrain—lakes, valley floors, and lowland plains within the padded extent—therefore dominates this figure, which is why it far exceeds the count of strict, isolated summits. Each such pixel seeds a Union-Find component whose key col is assigned at its first qualifying merge (Theorem 1); equal-elevation seeds receive prominence 0 at their first tie merge and are removed by the subsequent ρ threshold, leaving 2677 raw peaks at $\rho \geq 130$ m. The global maximum never merges with higher terrain, which accounts for exactly one fewer key-col assignment than seeds. The subsequent stages (Steps 3–5d and the Step 4b export) completed successfully on the resulting national layer stack.

Table V
PixelMinimax performance on the national Austrian run (10 m resolution, 40 km cross-border padding).

Quantity	Value
Valid DEM pixels (n)	2 450 224 128
Local maxima (non-strict, incl. ties)	910 479 363
Components assigned a key col	910 479 362
Raw peaks ($\rho \geq 130$ m)	2677
External merge-sort runs	123
Union-Find index type	`int64` (SSD mmap)
Workstation memory	64 GB unified
SSD scratch footprint	95.8 GB
Step 2 wall-clock runtime	10 h 29 m 42 s

b) Interpretation: The out-of-core implementation trades lower peak RAM consumption for higher SSD traffic and a longer national Step 2 wall-clock time. This is acceptable because it preserves the seamless, non-tiled formulation of the prominence computation while remaining executable on a 64 GB workstation.

c) Optimisations Evaluated and Rejected: Several further Step 2 optimisations were tested on the production hardware (Apple Silicon, unified memory) and discarded because they gave no net benefit on this architecture. Union-Find path-halving was slower than the two-pass variant, which retains an L1-cache advantage and ran 30–40 % faster in practice. An in-memory bit-array for the processed-pixel flag was rejected because Python bit operations cost about 4× more than a direct `uint8` memory-mapped access. An in-loop `target_pk_idx` memory map for routing targets was also rejected: the additional ≈20 GB overhead outweighed all gains, which is why routing targets are instead resolved in a fast post-processing pass (Step 2b, `find_routing_targets.py`), which adds the fields `target_pk_x`, `target_pk_y`, `target_pk_elev`, and `target_pk_dist` (key col to routing target, km) to `peaks_prom_raw.gpkg` in seconds. In contrast, increasing the external merge-sort chunk size (`SORT_CHUNK`) does help and is retained. These results indicate that on unified-memory hardware the dominant cost is memory bandwidth and scratch traffic rather than raw instruction count, so cache-friendly two-pass access patterns are preferable to algorithmic micro-optimisations.

d) Scaling to Larger Geographies and Future Work: Scaling beyond Austria remains feasible in principle but becomes increasingly sensitive to SSD throughput, random-access behaviour in the Union-Find stage, and the size of the external sort. For larger geographies, either a coarser base DEM or a correspondingly wider refinement band will be required to keep national runs practical. Two quantitative studies are left to future work. First, a *scaling benchmark* across nested extents (e.g. single Bundesland to national) would characterise empirically how wall-clock time, RAM, and scratch traffic grow with n, complementing the $O(n \log n)$ bound. Second, a *controlled comparison against divide-tree implementations*

Table VI
BEV-FIRST NAME-SOURCE REVIEW QUEUES IN THE V5 REVIEWER COPY.

Queue	**Rows**
BEV_AUTH_AVAILABLE	2247
N1_NO_AUTHORITATIVE_BEV_NAME_SOTA_DB_FALLBACK	89
N1_NO_AUTH.._SOTA_DB_FALLBACK_WITH_GEO_OR_REPLAY	40
N1_NO_AUTHORITATIVE_BEV_NAME_ONLY	54
N1_NO_AUTHORITATIVE_BEV_NAME_WITH_GEO_OR_REPLAY	3
Calculated summits without BEV name coverage	**186**

such as Kirmse's continental-scale tool [3], [4]—on a common extent and DEM—would quantify the relative trade-offs in runtime, memory, and localisation accuracy. Both require a harmonised experimental setup beyond the scope of the present national study.

Appendix B
Review-Layer Footprints

This appendix reports the full per-flag breakdown of the post-validation review layers (Steps 5c and 5d), which is summarised in the main results (Section IV-B). These figures are operational diagnostics; they do not affect any prominence, key-col, saddle, or matching value.

Step 5c (coordinate and summit-identity validation) checks three point sources—the SOTA/DB anchor, the BEV-NAMEN reference point, and the calculated DEM peak—against the ALS 1 m DEM. It preserves all 3831 input rows and adds diagnostics only. The all-row issue footprint is 911; the Austrian/non-foreign Table-V footprint is 899. Coordinate, identity, and name flags occur in 894, 49, and 208 rows, respectively.

Step 5d (coordinate-aware final assignment) assigns explicit roles to all 3831 rows. The exclusive final-role partition is 1849 `FINAL_SOTA_SUMMIT`, 275 `FINAL_SOTA_SUMMIT_REVIEW`, 324 `REVIEW_REQUIRED`, 71 `UNRESOLVED_REFERENCE`, 8 `SECONDARY_SHARED_KEYCOL`, 6 `REPLAY_REQUIRED`, and 1298 `FOREIGN_CONTEXT`. Consequently, `final_sota_flag`=2124=1849+275. Manual review is required for 681 rows overall and for 676 Austrian/non-foreign Table-V rows; `final_replay_required`=1 in 102 rows.

A. Name-Source and OpenStreetMap Review Layers

1) Name-Source Review Queue and SOTA-DB Fallback Names: The v5 freeze keeps a separate name-source review queue apart from the Step 5c coordinate/name/identity issue classes. This queue is not defined by whether a row has any display name. Instead, it is defined by whether the calculated summit has an authoritative BEV name at the calculated summit. A SOTA-DB name is retained as an existing database reference or fallback label, but it is not treated as proof of BEV name coverage and it must not suppress a name-source review case.

The reviewer copy therefore adds helper fields such as `bev_calc_name_available`, `sota_db_name_available`, `assigned_name_is_sota_fallback`, `osm_name_review_allowed`, and `name_source_review_queue`. For example, a row such as Rosalienkapelle can have a SOTA-DB fallback name but no BEV name at the calculated summit; it must remain visible as a name-source review case. The reviewer then records the accepted final name and its source in the Excel review sheet.

2) Optional OpenStreetMap Name Evidence: OpenStreetMap peak names are treated as a supplementary review layer, not as part of the productive v5 computation. The intended QGIS workflow is: load an OSM `natural=peak` layer, inspect candidate names around the N1 rows, and let the reviewer accept or reject candidate names in the review sheet. OSM must not overwrite BEV `F_CODE=7302` summit anchors or BEV `F_CODE=7303` secondary official name/context references. Accepted OSM names are database update decisions and are applied only in Future Step 6.

The name-source priority for the review workflow is therefore:

1) BEV `F_CODE=7302`: strict summit reference / stable summit anchor;
2) BEV `F_CODE=7303`: secondary official name/context reference;
3) BEV `F_CODE=7301`: terrain/context only;
4) SOTA-DB name: existing reference/fallback label, not BEV coverage;
5) OSM `natural=peak`: supplementary fallback evidence only if no BEV name exists;
6) final manual reviewer decision.

The reviewer-sheet guard is `osm_name_review_allowed=1`. A reviewer may choose `USE_OSM_PEAK_NAME` only in that case. The OSM candidate fields are `osm_candidate_name`, `osm_candidate_ele`, `osm_candidate_dist_m`, `osm_name_similarity`, and `osm_support_class`. They are review evidence fields only and do not affect prominence, key-col geometry, final roles, or replay flags.

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